Topics in Geometry and Topology:Assignment 5

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Problem 1

Show that any two-dimensional complete toric variety is projective. Show that any ample divisor on such a variety is very ample.

Solution: Let X_Σ be a two-dimensional complete toric variety. We first show that Σ admits a strictly convex integral support function. Order the rays cyclically as $\rho_i = \mathbb{R}_{\geq 0} v_i$, where $i = 0, \ldots, r$ and $v_i \in N$ is primitive. Put $\sigma_i = \langle v_i, v_{i+1} \rangle$, with indices taken modulo $r+1$. For each i, choose a primitive vector $u_i \in \rho_i^\perp \cap \sigma_i^\vee \cap M$, oriented so that it is positive on the interior of σ_i.

We claim that there exist positive integers $c_i > 0$ such that

$$\sum_{i=0}^{r} c_i u_i = 0.$$

Indeed, otherwise the vectors u_i would lie in a closed half-plane through the origin by the separating hyperplane theorem. This would imply that the rays ρ_i also lie in a closed half-plane, contrary to the completeness of Σ.

Choose $m_0 \in M$ and define $m_i = m_{i-1} + c_i u_i$. Define a piecewise linear function ψ by $\psi|_{\sigma_i} = \langle m_i, - \rangle$. Since $m_i - m_{i-1} = c_i u_i$ vanishes on ρ_i, the linear functions on σ_{i-1} and σ_i agree on their common ray. Hence ψ is continuous. Moreover, because $c_i > 0$ and u_i is positive on the interior of σ_i and negative on the interior of σ_{i-1}, the function ψ is strictly convex. Therefore X_Σ is projective.

Now let D be an ample divisor on X_Σ. Replacing D by a linearly equivalent torus-invariant Cartier divisor, write $D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$. Its associated polytope is

$$P_D = \{ m \in M_\mathbb{R} \mid \langle m, v_\rho \rangle \geq -a_\rho \text{ for all } \rho \in \Sigma(1) \}.$$

Since D is ample, P_D is a lattice polygon whose normal fan is Σ.

By the toric criterion for very ampleness, it is enough to show that for each maximal cone $\sigma \in \Sigma$, the semigroup $\sigma^\vee \cap M$ is generated by $(P_D \cap M) - m_\sigma$, where $m_\sigma \in P_D \cap M$ is the vertex corresponding to σ. Fix such a cone σ, translate by $-m_\sigma$, and set $Q = P_D - m_\sigma$. Then 0 is a vertex of Q, and the tangent cone of Q at 0 is $\sigma^\vee$.

Let $e_1, e_2 \in M$ be the primitive lattice directions of the two edges of Q starting at 0. Then $\sigma^\vee = \mathbb{R}_{\geq 0} e_1 + \mathbb{R}_{\geq 0} e_2$, and $e_1, e_2 \in Q \cap M$. The triangle with vertices $0, e_1, e_2$ lies in Q by convexity. List the primitive lattice vectors in this triangle in angular order as $w_0 = e_1, w_1, \ldots, w_s = e_2$.

For each j, the consecutive pair w_j, w_{j+1} forms a $\mathbb{Z}$-basis of M. Otherwise, the parallelogram spanned by w_j, w_{j+1} would contain a nonzero lattice point not generated by them, giving a lattice direction strictly between w_j and w_{j+1}, contradicting their consecutiveness. Hence every lattice point in $\mathbb{R}_{\geq 0}w_j + \mathbb{R}_{\geq 0}w_{j+1}$ is a non-negative integral combination of w_j and w_{j+1}. Since these cones cover $\sigma^\vee$, every element of $\sigma^\vee \cap M$ is generated by the w_j's. As each $w_j \in Q \cap M$, the semigroup $\sigma^\vee \cap M$ is generated by $Q \cap M = (P_D \cap M) - m_\sigma$.

Therefore the toric criterion for very ampleness holds for every maximal cone σ, and hence D is very ample. ■

Problem 2

If $X(\Delta)$ is complete and nonsingular, show that a T-divisor is ample if and only if it is very ample.

Solution: Since very ample divisors are ample, it remains to prove that every ample divisor is very ample.

Let $D = \sum_{\rho \in \Delta(1)} a_\rho D_\rho$ be an ample T-invariant divisor on $X(\Delta)$. Since $X(\Delta)$ is nonsingular, every T-invariant divisor is Cartier. For each maximal cone $\sigma \in \Delta$, let $m_\sigma \in M$ be the Cartier datum of D, so that $\langle m_\sigma, v_\rho \rangle = -a_\rho$ for every $\rho \subset \sigma$. Define

$$P_D = \{m \in M_\mathbb{R} \mid \langle m, v_\rho \rangle \geq -a_\rho \text{ for all } \rho \in \Delta(1)\}.$$

Since D is ample, m_σ is the vertex of P_D corresponding to σ.

By the toric criterion for very ampleness, it is enough to prove that, for every maximal cone σ, the semigroup $\sigma^\vee \cap M$ is generated by $(P_D \cap M) - m_\sigma$.

Fix a maximal cone $\sigma = \langle v_1, \ldots, v_n \rangle$. Since $X(\Delta)$ is nonsingular, $v_1, \ldots, v_n$ form a $\mathbb{Z}$-basis of N. Let $u_1, \ldots, u_n$ be the dual basis of M, so that $\langle u_i, v_j \rangle = \delta_{ij}$. Then $\sigma^\vee \cap M = \mathbb{Z}_{\geq 0}u_1 + \cdots + \mathbb{Z}_{\geq 0}u_n$. Thus it suffices to show that $u_i \in (P_D \cap M) - m_\sigma$ for every i.

Fix i, and let $\tau_i = \langle v_1, \ldots, \widehat{v_i}, \ldots, v_n \rangle$. Since $X(\Delta)$ is complete, there is a maximal cone $\sigma_i' \neq \sigma$ such that $\sigma \cap \sigma_i' = \tau_i$. Let $m_{\sigma_i'}$ be the Cartier datum of D on σ_i'. Since $m_{\sigma_i'}$ and m_σ agree on τ_i, we have $\langle m_{\sigma_i'} - m_\sigma, v_j \rangle = 0$ for all $j \neq i$. Hence $m_{\sigma_i'} - m_\sigma = c_i u_i$ for some integer c_i. Since D is ample, the support function is strictly convex, so $c_i > 0$.

Both m_σ and $m_{\sigma_i'}$ are vertices of the convex polytope P_D, and hence the line segment between them lies in P_D. Since

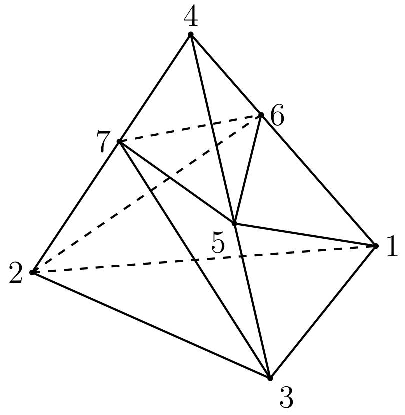
$$m_\sigma + u_i = \left(1 - \frac{1}{c_i}\right) m_\sigma + \frac{1}{c_i} m_{\sigma_i'},$$

we have $m_\sigma + u_i \in P_D \cap M$. Therefore $u_i = (m_\sigma + u_i) - m_\sigma \in (P_D \cap M) - m_\sigma$.

Thus $(P_D \cap M) - m_\sigma$ generates $\sigma^\vee \cap M$ for every maximal cone σ. By the toric criterion for very ampleness, D is very ample. ■

Problem 3

Form the three-dimensional fan Δ whose edges in $\mathbb{Z}^3$ pass through $v_1 = -e_1$, $v_2 = -e_2$, $v_3 = -e_3$, $v_4 = e_1 + e_2 + e_3$, $v_5 = v_3 + v_4$, $v_6 = v_1 + v_4$, and $v_7 = v_2 + v_4$ and with cones through the faces of the triangulated tetrahedron shown below:



Show that the toric variety associated to the fan above is complete and nonsingular, and determine whether it is projective or not.

Solution: It is complete and nonsingular but it is not projective. We use notation Δ_{ijk} to denote the triangulated tetrahedron generated by vertex v_i, v_j, v_k and origin. Let Δ_{0123} denote the triangulated tetrahedron generated by vertices v_1, v_2, v_3, v_4.

To see the completeness, The decomposition

$$\Delta_{0123} = \Delta_{123} \cup \Delta_{135} \cup \Delta_{156} \cup \Delta_{456} \cup \Delta_{237} \cup \Delta_{357} \cup \Delta_{457} \cup \Delta_{126} \cup \Delta_{267} \cup \Delta_{467}.$$

implies the completeness. Since it gives a subdivision of Δ_{0123}. Let Y be the toric variety corresponding to the fan generated by $\Delta_{123}, \Delta_{134}, \Delta_{234}, \Delta_{124}$. Clearly, Y is complete and $X(\Delta)$ as the blow-up of Y maintains the completeness.

Let determinant of v_i, v_j, v_k denoted by d_{ijk}. Then

$$d_{123} = d_{267} = d_{135} = d_{156} = d_{457} = -1 \qquad d_{126} = d_{467} = d_{456} = d_{237} = d_{357} = 1$$

implies the smoothness of $X(\Delta)$.

Finally, we show that $X(\Delta)$ is not projective. Suppose, for contradiction, that $X(\Delta)$ is projective. Then Δ admits a strictly convex support function ψ. Put

$$a_i = \psi(v_i).$$

Using the linear relations

$$v_1 + v_7 = v_2 + v_6, \qquad v_3 + v_6 = v_1 + v_5, \qquad v_2 + v_5 = v_3 + v_7,$$

and the strict convexity of ψ, we get

$$a_1 + a_7 < a_2 + a_6, \qquad a_3 + a_6 < a_1 + a_5, \qquad a_2 + a_5 < a_3 + a_7.$$

Adding these three inequalities gives

$$(a_1 + a_7) + (a_3 + a_6) + (a_2 + a_5) < (a_2 + a_6) + (a_1 + a_5) + (a_3 + a_7),$$

which is impossible, since the two sides are equal. Hence no strictly convex support function exists on Δ. Therefore $X(\Delta)$ is not projective. ∎

Problem 4

For the toric variety $X = \mathbb{P}^n$ with its divisors $D_0, \ldots, D_n$, verify directly that:

1. $D = \sum a_i D_i$ is generated by its sections $\iff$ Ψ_D is convex $\iff$ $a_0 + \cdots + a_n \geq 0$, and

2. D is ample $\iff$ Ψ_D is strictly convex $\iff$ $a_0 + \cdots + a_n > 0$.

Solution: (1) Let $X = \mathbb{P}^n$ with its standard fan. The rays are generated by $v_0 = -e_1 - \cdots - e_n$ and $v_i = e_i$ for $1 \leq i \leq n$. The maximal cone σ_i is generated by all v_j with $j \neq i$. Let $D = \sum_{i=0}^n a_i D_i$, and put $s = a_0 + \cdots + a_n$.

For the Cartier datum $m_i = m_{\sigma_i}$, we have $\langle m_i, v_j \rangle = -a_j$ for $j \neq i$. Hence $m_0 = (-a_1, \ldots, -a_n)$, and for $i \geq 1$,

$$m_i = (-a_1, \ldots, -a_{i-1}, s - a_i, -a_{i+1}, \ldots, -a_n).$$

Thus $m_i - m_0 = s e_i^*$.

Now

$$P_D = \{m \in M_{\mathbb{R}} \mid \langle m, v_i \rangle \geq -a_i \text{ for all } i\}.$$

The divisor D is generated by global sections iff $m_i \in P_D$ for every maximal cone σ_i. For m_0, the only inequality to check is the one for v_0, and it gives $s \geq 0$. For m_i with $i \geq 1$, the only inequality to check is the one for v_i, and it also gives $s \geq 0$. Therefore

$$D \text{ is generated by sections} \iff \Psi_D \text{ is convex} \iff s \geq 0.$$

(2) For strict convexity, the Cartier data on adjacent maximal cones must be distinct and bend strictly. Since $m_i - m_0 = s e_i^*$, this happens exactly when $s > 0$. Hence

$$\Psi_D \text{ is strictly convex} \iff s > 0.$$

Finally, since $D_i \sim H$ for all i, we have $D \sim sH$. Thus $\mathcal{O}(D) \simeq \mathcal{O}_{\mathbb{P}^n}(s)$, which is ample iff $s > 0$. Hence

$$D \text{ is ample} \iff \Psi_D \text{ is strictly convex} \iff a_0 + \cdots + a_n > 0.$$

∎

Problem 5

Let

$$A = \{P \subseteq M_{\mathbb{R}} \mid P \text{ is a full dimensional rational polytope}\},$$
$$B = \{(X(\Delta), D) \mid D \text{ is a } T_N\text{-invariant ample divisor in } X(\Delta)\},$$

where Δ is a complete fan, i.e. $|\Delta| = N_{\mathbb{R}}$. Show that the maps $P \mapsto (X_P, D_P)$ and $(X(\Delta), D) \mapsto P_D$ define bijections between the sets A and B which are inverses of each other.

Solution: Let $P \subset M_{\mathbb{R}}$ be a full-dimensional rational polytope, and let Δ_P be its normal fan. Since P is full-dimensional, Δ_P is complete. For every ray $\rho \in \Delta_P(1)$, let $v_\rho \in N$ be its primitive generator, and set $a_\rho = -\min_{m \in P}\langle m, v_\rho\rangle$. Define $D_P = \sum_{\rho \in \Delta_P(1)} a_\rho D_\rho$. Then

$$P = \{m \in M_{\mathbb{R}} \mid \langle m, v_\rho\rangle \geq -a_\rho \text{ for all } \rho \in \Delta_P(1)\}.$$

Indeed, these are precisely the facet inequalities of P, so $P_{D_P} = P$. Moreover, the support function $\Psi_P(v) = \min_{m \in P}\langle m, v\rangle$ is strictly convex with respect to the normal fan Δ_P. Hence D_P is ample on $X(\Delta_P)$, and the assignment $P \mapsto (X_P, D_P)$ is well-defined.

Conversely, let $(X(\Delta), D) \in B$, where $D = \sum_{\rho \in \Delta(1)} a_\rho D_\rho$ is an invariant ample divisor. Define

$$P_D = \{m \in M_{\mathbb{R}} \mid \langle m, v_\rho\rangle \geq -a_\rho \text{ for all } \rho \in \Delta(1)\}.$$

Since D is ample, its support function Ψ_D is strictly convex with respect to Δ. Therefore P_D is a full-dimensional rational polytope, and its normal fan is exactly Δ.

It remains to check that the two constructions are inverse. Starting from P, the definition of D_P gives $P_{D_P} = P$. Starting from $(X(\Delta), D)$, the ampleness of D implies that the normal fan of P_D is Δ. Moreover, for every ray $\rho \in \Delta(1)$, one has $\min_{m \in P_D}\langle m, v_\rho\rangle = -a_\rho$, since the inequality $\langle m, v_\rho\rangle \geq -a_\rho$ defines the corresponding facet of P_D. Hence

$$D_{P_D} = \sum_{\rho \in \Delta(1)} a_\rho D_\rho = D.$$

Thus the assignments $P \mapsto (X_P, D_P)$ and $(X(\Delta), D) \mapsto P_D$ are inverse bijections. ■